

K – means evaluation

There are two question we need to ask ourslef

- 1. How many clusters we need to choose*
- 2. is the observation belongs to grouped cluster or not*

*As we know alreday k = number is a hyper parameter
which gives the number of clusters*

- It is very important how to choose good clusters*
- Choosing maximum clusters and choosing minimum clusters not an effective way*
- we need to choose optimal clusters*

1) Within cluster sum of squares (WCSS)

- elbow method*
- 2) silhouette score*
 - 3) Heirarchical cluster*

Imagine that we have 10 observations

How many clusters are possible : 10 clusters

- Every observations consider as one group*

How many minimum clusters: 1 cluster

- All the observations together as one cluster*

*1 cluster is not good
spread ness is more
the observations are not tightly together
the distance between each observation to the means point is more*

Id	X	Y
1	1	2
2	2	3
3	3	4
4	4	5
5	5	6
6	6	7
7	7	8
8	8	9
9	9	10
10	10	11
11	11	12
12	12	13

For $k = 2$

Total data points are 12 divided into two clusters

First 6 points are one clusters ,second 6 points are another cluster

$$d_{1c1} + d_{2c1} + d_{3c1} + d_{4c1} + d_{5c1} + d_{6c1} + d_{7c2} + d_{8c2} + d_{9c2} + d_{10c2} + d_{11c2} + d_{12c2}$$

First we are choosing the number of cluster

then we are rearranging the observations

Our main : Are we choosing number of clusters correctly or not

suppose we choose $k = 2$,2 clusters

first 6 observations in one cluster

second 6 observations in second cluster

so we are calculating distance between each observation with same cluster mean

$$d_{1c1}, d_{2c1}, d_{3c1}, d_{4c1}, d_{5c1}, d_{6c1}$$

$$d_{7c2}, d_{8c2}, d_{9c2}, d_{10c2}, d_{11c2}, d_{12c2}$$

Now add all the distances

$$d_{1c1} + d_{2c1} + d_{3c1} + d_{4c1} + d_{5c1} + d_{6c1} + d_{7c2} + d_{8c2} + d_{9c2} + d_{10c2} + d_{11c2} + d_{12c2} = WCSS(2)$$

K	wcss
1	W1=5000
2	W2= 2000

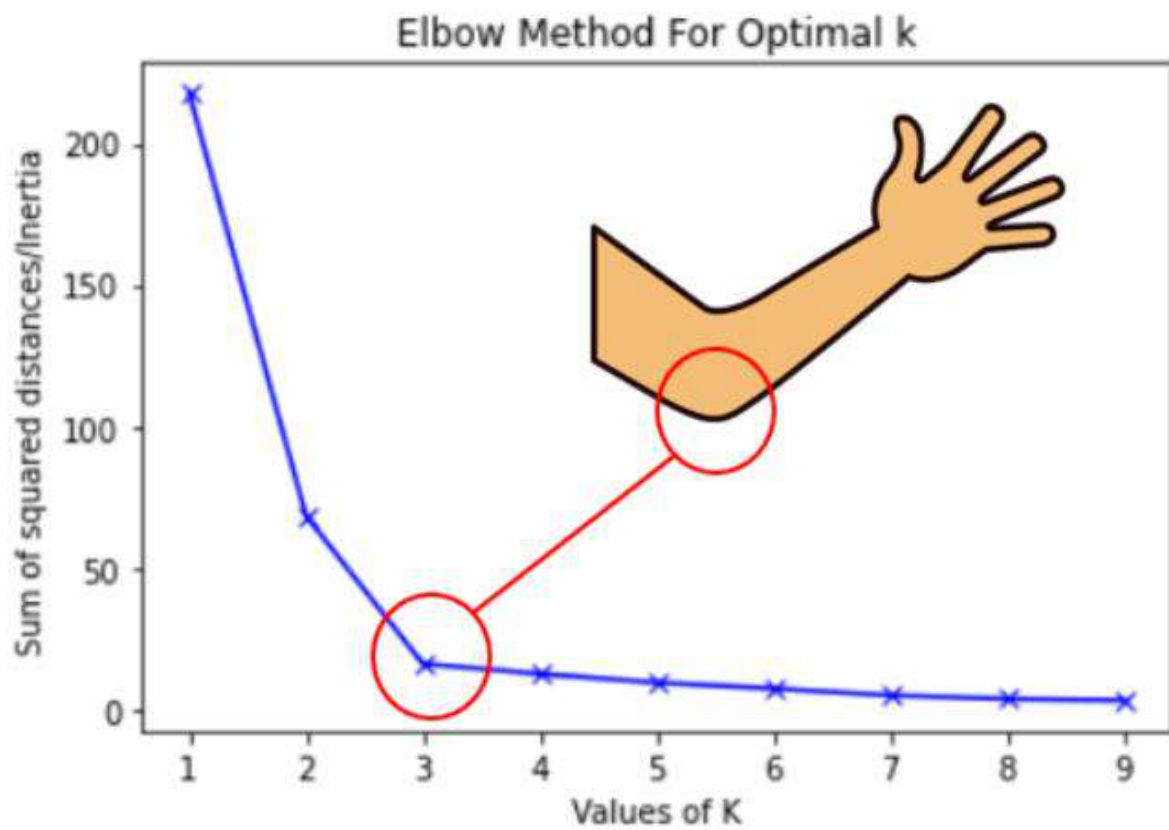
3	W3=1000
4	W4=700
5	W5=600
6	W6=550
7	W7=520

$W_1 > W_2 > W_3 > W_4 > W_5 > W_6 > W_7$

The deviation between w_1 and w_2 is so high

Compare to w_2 and w_3

After some clusters w_6 is almost equal to w_7



Line plot between K and inertia

Id	X	Y
1	1	2
2	2	3
3	3	4
4	4	5
5	5	6
6	6	7

7	7	8
8	8	9
9	9	10
10	10	11
11	11	12
12	12	13

$k = 1$ clustr
 $k = 2 = (1,2,3,4,5,6) (7,8,9,10,11,12)$
 $k = 3 (1,2,3,4) (5,6,7,8) (9,10,11,12)$

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$k = 1$
 $[(1,2), (2,3), (3,4), (4,5), (5,6), (6,7), (7,8), (8,9), (9,10), (10,11), (11,12), (12,13)]$
 $centroid = (6.57, 7.5) \text{ WCSS} = 286$

$k = 2$
 $[(1,2), (2,3), (3,4), (4,5), (5,6), (6,7),] \text{ centroid} = (3.5, 4.5) \text{ WCSS} = 35$
 $[(7,8), (8,9), (9,10), (10,11), (11,12), (12,13)] \text{ centroid} = (9.5, 10.5) \text{ WCSS} = 35$

$k = 3$
 $[(1,2), (2,3), (3,4), (4,5)], \text{ centroid} = (2.5, 3.5) \text{ WCSS} = 10$
 $[(5,6), (6,7), (7,8), (8,9)] \text{ centroid} = (6.5, 7.5) \text{ WCSS} = 10$
 $[(9,10), (10,11), (11,12), (12,13)] \text{ centroid} = (10.5, 11.5) \text{ WCSS} = 10$

k	Type equation here.	WCSS
1	286	286
2	35 + 35	70
3	10 + 10 + 10	30
4(assume)		we wil get value less than 30 = 27

Conclusion: The WCSS disfference between $k = 1$ to $k = 2$ is maximum
 $k = 2$ to $k = 3$ modertae deviation
 $k = 3$ to $k = 4$ very less deviation
then we can choose $k = 3$ optimal cluster

WCSS also called as Intertia
In python the method name is Inertia